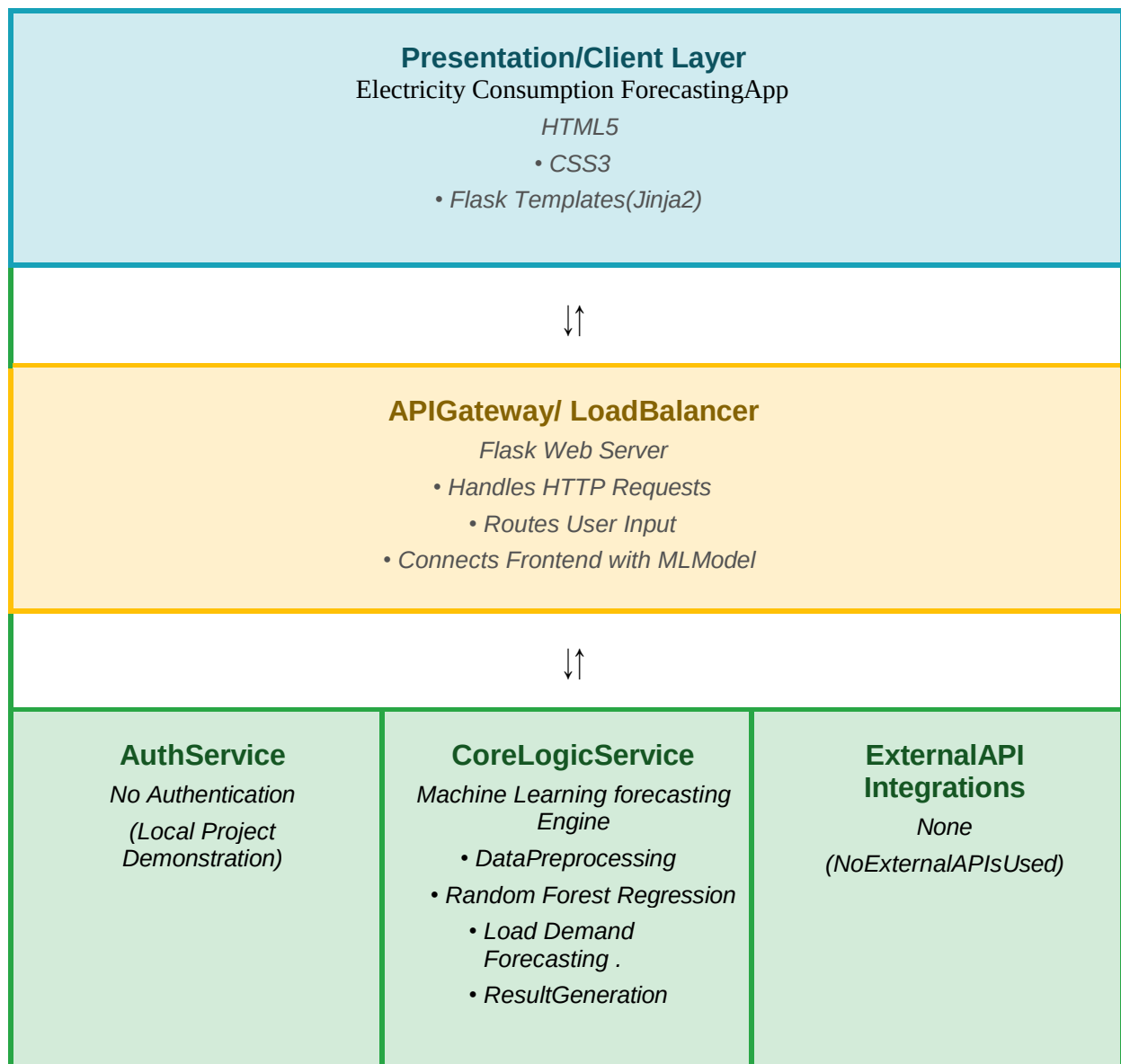


SolutionArchitecture

Date	6July2026
Team ID	
ProjectName	Plugging into the Future : An Exploration of Electricity consumption Patterns
MaximumMarks	5Marks

Solution Architecture Diagram:





Data/StorageLayer

Database / Storage Layer
MySQL / PostgreSQL / SQLite
Stores Historic Consumption Data
User Logs / Application Data

ComponentDescriptionTable

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a web interface for entering applicant information and displaying prediction results.	HTML5,CSS3,Flask Templates
API Gateway	Receives user input ,processes requests,loads the trained model, and generates predictions.	Python, Flask
Micro services	Performs preprocessing and predicts whether the applicant is a Good Customer or Bad Customer using the trained Random Forest model.	Scikit-learn,NumPy, Pandas
Database	Stores data sets used for training and the serialized machine learning model.	CSVFiles,Pickle (.pkl)